Ugly Numbers

Different numbers evoke different images to you. 4 is timid and shy, 24 is bold and aggressive, 25 is aloof and mysterious, and of course, any number that ends in a 9 is ugly.

The *ugliness* of a number is the number of trailing 9's it has. For example, 96 has ugliness 0, 9199 has ugliness 2, and 69999 has ugliness 4.

`http://orac.amt.edu.au/april/team/nine.txt` contains a long number, written digit by digit. What is its ugliness?

Ooh, Shiny

You are reading a list of numbers (which you can download from `http://orac.amt.edu.au/april/team/important.txt`) one by one. As you read the first number, your boredom is zero. For each successive number:

- If that number is the same as the previous number, your boredom increases by one.
- If that number is different to the previous one, your boredom becomes zero again.

What number are you reading as you experience maximum boredom?

Septic Septets

Oh no! You are trapped in a sewer! Your only hope is to read through the integers in `http://orac.amt.edu.au/april/team/sewer.txt` and identify all the multiples of seven!

...wait, what?

Atlantis XLII

The file `http://orac.amt.edu.au/april/team/water.txt` gives a grid of heights.

Determine the number of flooded cells in the map. (A cell is *flooded* if its height is strictly below forty-two.) You are guaranteed that all height values will be positive and odd. Negative values represent wells, and even values represent geysers. Neither of these will appear in the input data. All the squares surrounding a well are also considered flooded. A geyser inverts the floodedness of all surrounding squares.

Quest for Gold

You are exploring the Andes mountain range. The Andes can be represented as a series of altitudes. These are supplied in `http://orac.amt.edu.au/april/team/andes.txt`.

A *peak* of width n is a series of altitudes $\{a_i\}$ such that $a_1 < a_2 < ...a_{M-1} < a_M > a_{M+1} > ... > a_{n-1} > a_n$, with $1 \leq M \leq n$. Find the maximum width of any peak in the Andes.

Mystery Number

Find the smallest positive integer such that:

- Its remainder when divided by 31 is 15.
- Its remainder when divided by 37 is 36.
- Its remainder when divided by 901 is 18.

Where's Wally?

You are given a grid of letters, and told that the word '`WALLY`' appears somewhere in this grid, going any of eight directions.

```
QXPYFDS
FMXLSAF
DOWLLDA
FDAAFMW
MSAWDSY
IIOPESN
```

In the above grid, the word '`WALLY`' begins at row 5, column 4, and goes directly up.

`http://orac.amt.edu.au/april/team/thereheis.txt` contains another grid to solve. Determine the row and column where '`WALLY`' begins.

Palindromes

Palindromes are positive integers that read the same forwards as backwards. For example, 343 and 1001 are palindromes. 010 is not, as we cannot write a number beginning with a zero.

What is the 1500th palindrome?

Bugs II

Bugs invade your town. They swarm around for one day. Then they are gone for two days. Then they are back for three days. Then they are gone for four days. How many of the first eight thousand days are bug free?

Concert Halls

You are responsible for hiring out seats to concert halls. Concert hall i has r_i rows and s_i seats per row, and you have sold t_i tickets for that concert hall. It turns out that due to a bug in your booking systems, some concert halls have been overbooked. You must refund tickets until the halls run out of happiness.
`http://orac.amt.edu.au/april/team/sitting.txt` contains a list of concert hall data given as triples '$r_i\, s_i\, t_i$'. Calculate the total number of refunds you must give.

Five Tutors

There are five informatics tutors. Each tutor hates a different informatics topic, and each tutor has a different favourite thing. For instance:

- The person who hates computational geometry likes turtles.
- Luke hates graph theory.
- The person who hates dynamic programming likes mudkips.
- Ben likes cheese.
- The person who hates data structures likes waffles.
- Jarrah likes turtles.
- Robert hates dynamic programming.

The person who likes spam – what informatics topic do they hate?

Who hates mathematics?

What is Josh's favourite thing?

Frowning Furiously

Read the following rules for deciding whether a bitmap of a face given with '`.`'s and '`X`'s is *smiling*.

A face consists of a number of face elements (FE): a face boundary, two eyes, a nose, a mouth and a number of miscellaneous elements (eyebrows, etc.). Each FE is a contiguous area of '`X`'s (using 8-neighbourhood). The face always contains a face boundary, two eyes and a mouth. All other FEs are optional.

The face boundary is an FE containing all other FEs. Each face has exactly one face boundary. The face boundary surrounds one or more areas of '`.`'s (using 4-neighbourhood), and all other FEs constituting this face are located within these areas.

Eyes always surround an area of '`.`'s (this area may contain other FEs). The mouth may or may not surround an area of '`.`'s, but no other FE surrounds an area of '`.`'s.

The mouth is always below the eyes (i.e., the bottommost '`X`' of the mouth is always below the bottommost '`X`' of any eye). The mouth has the greatest left-to-right length of all FEs below the eyes except for the face boundary.
A face is smiling if either the leftmost or rightmost '`X`' of the mouth is at the top of the mouth (i.e., there is no part of the mouth higher than it).

Go to `http://orac.amt.edu.au/april/team/grr.html` and download the bitmaps linked to there. Which ones are smiling?

Max-Max

Max-Max starts at the top-left corner of a matrix of numbers. He moves to the bottom-right corner using only downward and rightward steps. What is the maximum possible sum of the numbers he walks across (including the corners)?

To assist you, you are told what the matrix of numbers is. Go to `http://orac.amt.edu.au/april/team/matrix.txt`.

Curry

You have c portions of curry, r portions of rice, and v portions of vegetables. The food is so unbearable that the only way you can eat is to eat one portion of two different food types at once. (For example, you may eat one portion of rice with one portion of curry. You may not eat two portions of vegetables, or one portion of vegetables by itself.)
Depending on your starting values, you may not be able to finish your entire meal. Download the list of scenarios at `http://orac.amt.edu.au/april/team/masala.txt` and find out how many of these make it possible for you to finish your meal.

Om Nom

The townsfolk of North Yorkshire have 100 hams. They decide to place these hams onto a line of plates so that the hippos may feast.

- They may use any number of plates.
- They must use all the hams.
- Each plate must have at least one ham. (Don't patronise a hippo.)
- No plate can have more hams than the plate right before it.

How many ways can they place hams onto plates so that these conditions are satisfied?

PHUN

You are organising this camp's PHUN. You must split the fifteen other students into groups of 1, 2, 3, 4 and 5 (one of each). How many ways can you do this?

Laser Tag

The chance of an informatics student getting shot during a laser tag game is 97%. How many informatics students should play laser tag to maximise the probability that, at the end of the game, exactly one student has not been shot?

Exciting Circles

Greg writes down a list of positive integers, and gives Samuel the following challenge:

> "Circle groups of three numbers, without overlapping. For each circle, add the value of the outermost two to your score. For example, if you draw circles where indicated around the list '`1 2 [3 4 5] 6 [7 8 9]`', your total score is $(3 + 5) + (7 + 9) = 24$."

Samuel repeatedly circles the group of three numbers which increases his score by the highest amount, until he cannot circle any more groups.

Find a list where this will *definitely* not give Samuel the optimal score. This list should be as small as possible. If there is more than one such list, output the earliest in lexicographical order.

Loch Ness

The Loch Ness monster is hiding inside a 1000×1000 grid of squares. For a million dollars per go, you can perform a *monster hunt*:

- Select a subgrid.
- You find out whether the monster is in this subgrid.
- If the monster is in this subgrid, and your subgrid is 1×1, you catch the monster.

How much money do you need to catch the Loch Ness monster?

Bagpipes

Your sister is learning to play the bagpipes, and you are having trouble finding a quiet spot in the house.

The file `http://orac.amt.edu.au/april/team/house.txt` describes your house as a rectangular grid of rooms (‘`.`’) and walls (‘`#`’). Your sister's room is marked with an ‘`X`’. Two rooms are adjacent if they share a common edge. Of course, any room can be reached from any other.

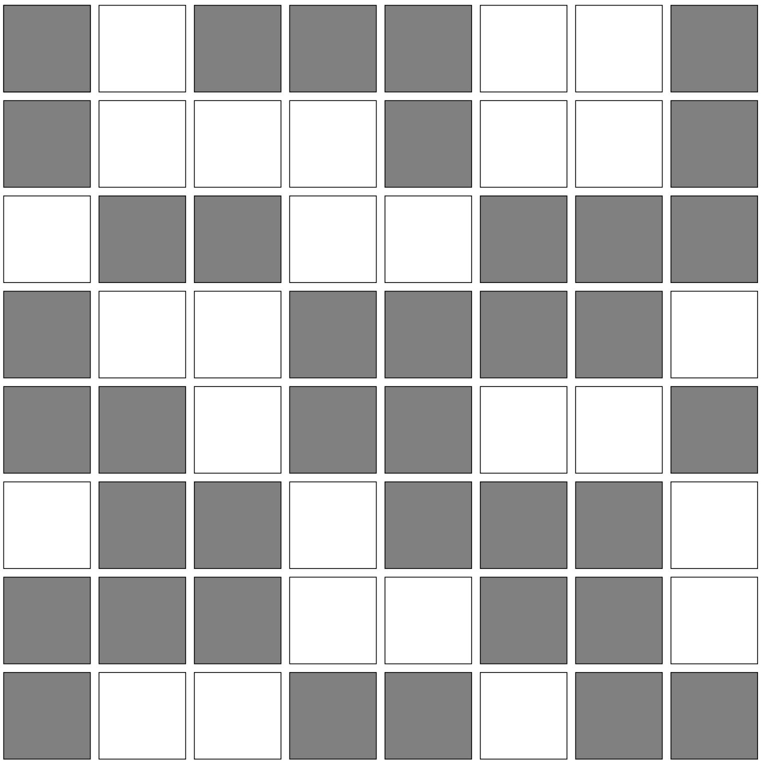
The *quietness* of a room is defined as:

- 0, if it is your sister's room
- min(quietness of adjacent rooms) + 1, otherwise

What is the highest quietness of any room in your house?

Chessmaster

Select exactly one square from each row and column of this chessboard. You may not select shaded squares.



Polygon

`http://orac.amt.edu.au/april/team/shape.txt` contains a list of points. Find the polygon with minimum perimeter that contains all the points (either in its interior or on the border). How many sides does that polygon have?

Mafia

Mafia is a popular game of chance. At the beginning of the game, players are assigned the role of Citizen or Mafia. The following procedure is then repeated until one of the groups wins:

- Kill a random player.
- If there are no Mafia players left, the Citizens win.
- Kill a random Citizen.
- If there are at least as many Mafia left as Citizens, the Mafia win.

Say that there are five players. Doing some probability analysis, we see that if there is 1 Mafia player to begin with, the Mafia has a $\frac{8}{15}$ chance of winning. If there are 2 Mafia players to begin with, the Mafia has a $\frac{13}{15}$ chance of winning.

Now say that there are 345 players. How many Mafia should the game begin with if we want the Mafia to have as close to a 50% chance of winning as possible?

The FARIO Ultimatum

You are a suave secret agent working for the Australian government. Your target: the clandestine FARIO organization, which has links to every black market in the free world. Your current mission: break into their Macquarie headquarters and discover what they are plotting.

You are faced with a keypad containing the digits 1 to 9. You know the secret password is up to 3 keys long. Find a minimal-length sequence of key presses such that every possible secret password is contained in the sequence.

(For example, if there were only three digits and passwords up to two digits long, you might use `1122331321`.)

Go to `http://orac.amt.edu.au/april/team/topsecret.php` and crack the keypad!

The Erdös Discrepancy

Alice and Bob are playing a game. First, Alice picks a number N, then Bob writes out a sequence of N integers, which are either 1 or -1. Alice then picks a number X and adds up the numbers at all the locations which are multiples of X.

For instance if Alice chooses $X = 5$ she would add up the numbers at positions $5, 10, 15, \ldots$. If the running total for this is outside the range $[-k, k]$ at any point, Alice wins! For some value of k at any rate....

If $k = 1$ Alice wins if she picks any $N \geq 12$. For Alice to win with $k = 2$, what's the smallest N she can pick?

Dvorak

James and Jarrah use Dvorak. They type ”`dvorak is great`” into a keyboard that has been set to QWERTY, and the computer shows ”`h.soav g; uodak`”. They then type ”`h.soav g; uodak`” in and the computer shows ”`je;sa.  uz fshav`”. How many *more* times do they have to type in the computer's output until the output is ¨`dvorak is great`”?

X Factor

Do you have the X Factor? You have the X Factor if you are an integer less than 10 million with exactly 10 factors. How many numbers could you be if you are confident that you have the X factor?

... And It's My Day

Name the song.

- There's a ___________ disc CD changer in her car
- It's 3 ___________ and here I come, so you should probably run
- ___________ the next day in breathing machines
- lose yourself ___________ music
- The ___________ rain clouds up my window

Algorithmic Verification

This is a bonus problem. If you solve this problem you may nominate one square of your choice (excluding centre squares), and get the tile for that square.

Write a rap battle between two competing algorithms or data structures. (e.g. 'array vs linked list' or 'Mergesort vs Quicksort').

- Each rap verse must have at least 8 lines.
- There must be rhymes between lines! (although rap rhyming is very flexible)
- Two members of the group must perform the rap battle.

Credit for this problem is awarded at the tutors' discretion!

Dynamic Programming

This is a bonus problem. If you solve this problem you may nominate one square of your choice (excluding centre squares), and get the tile for that square.

Draw Charlie the Wonderbudgie in his ski gear.

- Draw good.
- We can tell if you're not trying.
- "`xkcd` looks like that" is not an excuse.

Credit for this problem is awarded at the tutors' discretion!

Online Minimum Interval Cover

This is a bonus problem. If you solve this problem you may nominate one square of your choice (excluding centre squares), and get the tile for that square.

Explain a solution for *Elephants* through interpretive dance.

- All team members must be involved.
- *All.* Nobody is allowed to just stand by the side and narrate.
- The more amused we are at your expense, the better.

Credit for this problem is awarded at the tutors' discretion!